Urshita Pal

University of Michigan
Department of Mathematics
530 Church Street
Ann Arbor, MI, USA

urshita@umich.edu
urshitapal.github.io

EDUCATION

University of Michigan, USA
Ph.D., Mathematics, Expected 2027.
Advisor: Jennifer Wilson

University of Michigan, USA
M.S., Mathematics, 2021-23.

Chennai Mathematical Institute, India
B.Sc., Mathematics & Computer Science, 2018-21.

RESEARCH INTERESTS

(Co)homology of Arithmetic Groups, (Co)homological stability phenomena

AWARDS AND FELLOWSHIPS

Rackham Predoctoral Fellowship
Research Fellowship for the academic year 2025-26

Rackham One-Term Dissertation Fellowship
Research Fellowship for the academic term of Jan-April 2025

Gold Medal of Excellence, BSc Math & Computer Science
Chennai Mathematical Institute, 2021
Highest GPA at time of graduation

Bronze Medal, European Girls Math Olympiad
Held in Florence, Italy, 2018

International Math Olympiad Training Camp, Mumbai, India
Selected for Participation, 2017 & 2018

RESEARCH PAPERS

The Quillen and sharbly Hopf algebra structures on Steinberg homology coincide
with S. Payne
Arxiv preprint Arxiv:2609.38735

A projective resolution of the symplectic Steinberg module
Arxiv preprint Arxiv:2605.06499

The top cohomology of principal congruence subgroups of special linear groups over Euclidean number rings
Arxiv preprint Arxiv:2605.05087

Representation stability in the (co)homology of vertical configuration spaces
with D. Baron, C. Wang, J. Wilson, and C. Yang
ArXiv preprint ArXiv:2412.01128

TEACHING

University of Michigan
- Primary Instructor, Math 105 (Data, Graphs, & Functions)
Fall 2024 and Fall 2026
- Primary Instructor, Math 115 (Calculus 1)
Fall 2021, Winter 2022, Fall 2022, Winter 2023, and Fall 2023
(Duties included lecturing and facilitating groupwork for a class of 20-24 students three times a week, holding office hours, writing quizzes, and grading exams, quizzes, and problem sets)

Chennai Mathematical Institute
- Teaching Assistant
NPTEL online course on Rings & Fields, Jan-March 2021
- Teaching Assistant
Probability Theory, Spring 2020; Analysis 3, Fall 2020, Complex Analysis, Spring 2021; Topology, Spring 2021
(Duties included grading problem sets and holding office hours)

SERVICE AND MENTORSHIP

Michigan REU (Research Experience for Undergraduates), Summer 2024
-*Graduate Student Mentor for the REU hosted by Jenny Wilson*
(Duties included creating math presentations for undergraduate participants, mentoring undergraduate research, helping students present and write on mathematical projects.)

Michigan Directed Reading Program
-*Mentor in Winter 2023, Winter 2025, and Fall 2025*
(Duties included meeting one-on-one with students to discuss mathematical material, and helping students present and write on mathematical projects.)

Lab of Geometry at Michigan
-*Served on the Admissions Committee in Fall '23 & Winter '24*
(This is a semester-long program for providing mathematical research and computing experience for undergraduates.)

Graduate Student Dynamics/Geometry/Topology Seminar
-*Co-organiser 2022-23, Organiser 2023-24*

AWM (Association for Women in Math) Mentoring Programs
-*Mentor in the Mentor-Mentee program for the Academic Year 2022-23*
-*Participated in the 'Vertical Mentoring Program' in Winter 2024*

Michigan Math Club
-*Fagnano's Problem and Reflecting Triangles*, Jan 2023

TALKS

Upcoming Talks

-Joint Mathematics Meetings, January 2027

-Purdue University, December 2026

Invited Talks

-*The top cohomology of principal congruence subgroups of SL_nR and $Sp_{2n}R$*
Binghamton University, April 2026

-*The top cohomology of principal congruence subgroups of SL_nR and $Sp_{2n}R$*
Spring Redbud Topology Conference, University of Oklahoma, April 2026

-*Representation stability in the (co)homology of vertical configuration spaces*
Special Session at AMS Spring Sectional, Boston College, March 2026

-*The generalized Lee–Szczarba conjecture on the cohomology of principal congruence subgroups*
UC San Diego, February 2026

-*A Resolution of the Symplectic Steinberg Module*
Purdue University, October 2025

-*Steinberg Modules and Hopf Algebras* (expository)
University of Pennsylvania, March 2025

-*Steinberg Modules and $H^*(SL_n\mathbb{Z};\mathbb{Q})$* (expository)
Scissors Congruences, Algebraic K-Theory and Steinberg Modules, July 2024
(Held at the American Institute of Mathematics, Pasedena)

Contributed Talks

-*The top cohomology of congruence subgroups of SL_nR and $Sp_{2n}R$*
Algebraic Structures in Topology, San Juan, Puerto Rico, July 2026

-*Connectivity and Cohomology* (5 min lightning talk)
Young Geometric Group Theory, Bristol, April 2024

Summer Workshops

-*Segal's proof of homological stability* (expository)
Configuration Spaces Summer School, University of Michigan, May 2026

-*High Connectivity of Arc Complexes* (expository)
Talbot Workshop on Homological Stability, May 2025

CONFERENCES ATTENDED

- Algebraic Structures in Topology, San Juan, Puerto Rico, July 2026

- Configuration Spaces Summer School, University of Michigan, May 2026

- Spring Redbud Topology Conference, University of Oklahoma, April 2026

- Perspectives on the Cohomology of $\mathrm{GL}_n\mathbb{Z}$, Muenster, Germany, October 2025

- Scissors Congruences and Algebraic K-Theory, UPenn, July 2025

- Discrete Groups in Topology & Algebraic Geometry, Notre Dame, June 2025

- Polylogarithms, Homology of linear groups, & Steinberg Modules, June 2025
Held at the SwissMAP Research Station, Les Diablarets, Switzerland

- Talbot Workshop on Homological Stability, Cassapolis MI, May 2025

- Scissors Congruences, Algebraic K-Theory and Steinberg Modules, July 2024
Organised through the American Institute of Mathematics, Pasedena

- Young Geometric Group Theory XII, Bristol, April 2024

- Stability in Topology, Arithmetic and Representation Theory, Purdue, 2023
(Attended Virtually)

- Nearly Carbon Neutral Geometric Topology Conference, June 2023
(Attended Topic Group: Profinite and Residual Methods in Geometric Group Theory)

- Nearly Carbon Neutral Geometric Topology Conference, September 2022
(Attended Topic Group: Group Actions on Hyperbolic Spaces)